

WORLD CUP



Analysis of National Football Teams

PROBLEM

This project assumes we are part of the technical staff of the Mexican National Team.

Our goal is to analyze real data from rival national teams in this competition, in order to develop the best possible strategies to help the Mexican National Team reach the final stages of the 2026 World Cup.



SOLUTION

Develop a data engineering system that collects and processes information from international matches, generating performance metrics to support the technical staff's decision-making.



ARCHITECTURE

Technologies Used



Amazon EC2



Amazon S3



AWS Glue



Amazon Athena

INGESTION

DATA SOURCE



API-FOOTBALL
by api-sports.io



AMAZON EC2

Running Python
scripts to extract
data



STORAGE AND PROCESSING



BRONZE
(RAW)

Raw CSV data
/bronze/
(by selection /
date)



AWS GLUE
(SPARK)

Data
transformation
and cleaning



SILVER
(PROCESSED)

Clean and
structured
Parquet data
/silver/
(partitioned
by year)



AMAZON
ATHENA

SQL queries
on processed
data

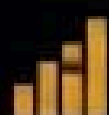


GOLD
(ANALYTICAL)

Aggregated and
ready-to-analyze
/gold/



GOLD LAYER - CONSUMPTION



METRICS AND TABLES

Ready-for-analysis performance
metrics

- Win Rate
- Goals For / Goals Against
- Performance by Opponent
- Performance by Tournament
- Yearly Trends

DATA QUALITY RULES



Null
Removal



Goal Range
Validation
(0-20)



Team Name
Standardization



Duplicate
Removal

RELIABLE, CONSISTENT, AND READY-FOR-ANALYSIS DATA

ATHENA QUERY EXAMPLE

```
SELECT m.tm_season, COUNT(+) AS matches,  
       AVG(m.goals_for) AS avg_goals_for  
FROM matches AS m  
WHERE m.season = '2023'  
GROUP BY m.team, m.season  
ORDER BY avg_goals_for DESC;
```


DATA INGESTION

- Data is retrieved from API-Football using Python scripts running on Amazon EC2.
- Data covers Mexico's future rivals and potential opponents.
- Matches and tournaments played over the last 4 years are extracted (August 2022 to May 2026).
- A separate CSV file is generated for each team's historical record.
- Files are stored in Amazon S3 — Bronze Layer.



ON THIS DAY ARGENTINE 3-3 FRANCE



KEY DATA FIELDS

- Match date
- National teams (home and away)
- Results — August 2022 to May 2026
- Competition / Tournament
- Goals (home and away)



BRONZE LAYER

2023-06-26	Mexico	Honduras	4	0	CONCACAF Gold Cup
2023-06-30	Haiti	Mexico	1	3	CONCACAF Gold Cup
2023-07-03	Mexico	Qatar	0	1	CONCACAF Gold Cup
2023-06-08	Mexico	Guatemala	2	0	Friendlies
2023-06-11	Mexico	Cameroon	2	2	Friendlies
2023-07-05	Mexico	Costa Rica	2	0	CONCACAF Gold Cup
2023-07-13	Jamaica	Mexico	0	3	CONCACAF Gold Cup
2023-07-16	Mexico	Panama	1	0	CONCACAF Gold Cup
2023-09-10	Mexico	Australia	2	2	Friendlies
2023-09-12	Mexico	Uzbekistan	3	3	Friendlies
2023-10-15	Mexico	Ghana	2	0	Friendlies
2023-10-18	Mexico	Germany	2	2	Friendlies
2023-12-17	Mexico	Colombia	2	3	Friendlies
2023-11-18	Honduras	Mexico	2	0	CONCACAF Nations League
2023-11-22	Mexico	Honduras	2	0	CONCACAF Nations League
2024-03-22	Panama	Mexico	0	3	CONCACAF Nations League
2024-03-25	Mexico	USA	0	2	CONCACAF Nations League

2023-06-26	Mexico	Honduras	4	0	CONCACAF Gold Cup
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2023-07-05	Mexico	Costa Rica	2	0	CONCACAF Gold Cup
2023-07-13	Jamaica	Mexico	0	3	CONCACAF Gold Cup
2023-07-16	Mexico	Panama	1	0	CONCACAF Gold Cup
2023-09-10	Mexico	Australia	2	2	Friendlies
2023-09-12	Mexico	Uzbekistan	3	3	Friendlies
2023-10-15	Mexico	Ghana	2	0	Friendlies
2023-10-18	Mexico	Germany	2	2	Friendlies
2023-12-17	Mexico	Colombia	2	3	Friendlies
2023-11-18	Honduras	Mexico	2	0	CONCACAF Nations League
2023-11-22	Mexico	Honduras	2	0	CONCACAF Nations League
2024-03-22	Panama	Mexico	0	3	CONCACAF Nations League
2024-03-25	Mexico	USA	0	2	CONCACAF Nations League

BRONZE TO SILVER

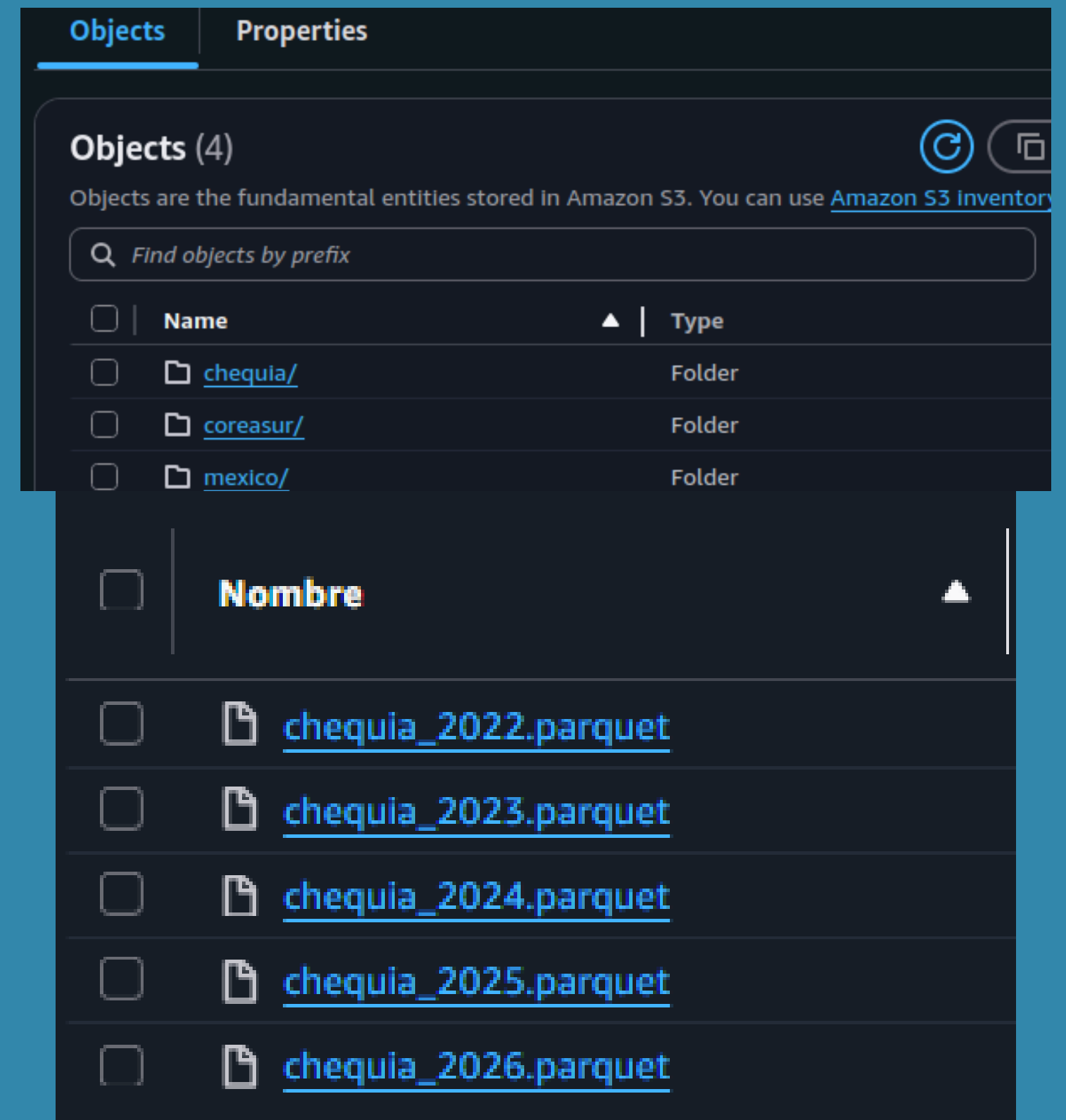
- Raw data stored in S3 is processed using AWS Glue with Apache Spark.
- The output is saved in Parquet format and data quality rules are applied.



SILVER LAYER

Data Quality Rules:

1. No null values in key columns.
2. Valid dates.
3. Goals between 0 and 20 for both home and away teams.
4. Home and away teams must be different.
5. Clean strings — no leading or trailing whitespace.
6. Standardized team names.
7. Matches split and partitioned by year.
8. HOME and AWAY status assigned to each participant.



SILVER LAYER DATA IN PARQUET FORMAT

aws

Buscar

[Alt+S]

Estados Unidos (Norte de Virginia)

hectordominguez (7596-2615-2179)

danieljose

Amazon Athena

Editor de consultas

Datos

Origen de datos

AwsDataCatalog

Catálogo

Ningún elemento

Base de datos

glue-2silver

Tablas y vistas

Crear

Filtrar tablas y vistas

Tablas (1)

2silver

fecha

timestamp

local

string

visitante

string

goles_local

int

goles_visitante

int

torneo

string

partition_0

string (Particionado)

Vistas (0)

Consulta 2

1 SELECT * FROM "2silver" LIMIT 10;

SQL Ln 1, Col 34

Ejecutar de nuevo

Explicar

Cancelar

Borrar

Crear

Volver a utilizar los resultados de la consulta hasta hace 60 minutos

Resultados de la consulta

Estado de la consulta

Completado

Tiempo en cola: 112 ms

Tiempo de ejecución: 947 ms

Datos analizados: 1.76 KB

Copiar

Descargar resultados en formato CSV

Resultados (10)

Filas de búsqueda

#	fecha	local	visitante	goles_local	goles_visitante	torneo	partition_0
1	2024-03-22 02:15:00.000	PANAMA	MEXICO	0	3	CONCACAF Nations League	mexico
2	2024-03-25 01:15:00.000	MEXICO	USA	0	2	CONCACAF Nations League	mexico
3	2024-06-01 01:30:00.000	MEXICO	BOLIVIA	1	0	Friendlies	mexico
4	2024-06-06 01:00:00.000	MEXICO	URUGUAY	0	4	Friendlies	mexico
5	2024-06-09 01:00:00.000	MEXICO	BRAZIL	2	3	Friendlies	mexico

CloudShell

Comentarios

Aplicación móvil de la consola

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SILVER TO GOLD

- Key metrics are generated using SQL queries in Amazon Athena over the structured Silver Layer data.
- The output is the final dataset ready for scouting analysis.

partidos ▼	victorias ▼	empates ▼	derrotas ▼
43	21	9	13

porcentaje_victorias ▼	goles_a_favor ▼	goles_en_contra ▼
48.84	66	44

goles_favor_por_partido ▼	goles_contra_por_partido ▼
1.53	1.02

GOLD LAYER



- Total matches
- Total wins
- Total draws
- Total losses
- Win percentage
- Total goals scored
- Total goals conceded

partidos ▾	victorias ▾	empates ▾	derrotas ▾
43	21	9	13

porcentaje_victorias ▾	goles_a_favor ▾	goles_en_contra ▾
48.84	66	44

goles_favor_por_partido ▾	goles_contra_por_partido
1.53	1.02

SCOUTING DASHBOARD (GOLD)

○ MEXICO NT — TECHNICAL STAFF

SCOUTING DASHBOARD

Historical data · Aug 2022 – present · Gold Layer

DATA LAKE / GOLD

OVERVIEW

HEAD-TO-HEAD

SELECT TEAM

MX MEXICO

KR SOUTH KOREA

ZA SOUTH AFRICA

CZ CZECH REPUBLIC

CA CANADA

QA QATAR

CH SWITZERLAND

BA BOSNIA

GB ENGLAND

HR CROATIA

DE GERMANY

SE SWEDEN

NO NORWAY

MATCHES PLAYED

43

Total historical

WINS

21

48.84% win rate

GOALS SCORED

66

1.53 per match

GOALS CONCEDED

44

1.02 per match

GOAL DIFF

+22

Overall

SCOUTING DASHBOARD (GOLD)

MX
MEXICO

4 ADVANTAGES

VS

ZA
SOUTH AFRICA

3 ADVANTAGES

METRIC DUEL

43

MATCHES PLAYED

47

▲ EDGE

21

WINS

19

▲ EDGE

48.84%

WIN RATE

40.43%

▲ EDGE

DOES THE GOLD LAYER DATA SOLVE THE PROBLEM?

Yes. The Gold Layer data allows the technical staff to analyze team performance and make informed, data-driven decisions.

The generated metrics enable evaluating the offensive and defensive performance of each national team, identifying patterns and tendencies across recent international results, comparing rival teams directly against Mexico across all key metrics and supporting strategic decision-making ahead of World Cup 2026 matchups.